Geometric Series Worksheet — Worked Answer Key

Twenty finite, infinite, shifted-index, decimal, modeling, and error-analysis problems.

Revision: 2026-07-23 **Canonical page:** https://bettergrades.net/subjects/math/calculus/worksheets/geometric-series/

1. Find the common ratio of 3, 12, 48, …

   Answer: $r = 4$

   a. Identify the first included term, common ratio, and whether the sum is finite or infinite.
   b. Use $S_n = a(1 - r^n)/(1 - r)$ for a finite sum or $S = a/(1 - r)$ only when $|r| < 1$.
   c. Substitution and simplification give $r = 4$.

2. Find a_8 when a_1=5 and r=2.

   Answer: $a_8 = 640$

   a. Identify the first included term, common ratio, and whether the sum is finite or infinite.
   b. Use $S_n = a(1 - r^n)/(1 - r)$ for a finite sum or $S = a/(1 - r)$ only when $|r| < 1$.
   c. Substitution and simplification give $a_8 = 640$.

3. Evaluate $\sum_{k=0}^{5} 3(2)^k$.

   Answer: 189

   a. Identify the first included term, common ratio, and whether the sum is finite or infinite.
   b. Use $S_n = a(1 - r^n)/(1 - r)$ for a finite sum or $S = a/(1 - r)$ only when $|r| < 1$.
   c. Substitution and simplification give 189.

4. Evaluate $\sum_{k=0}^{4} 16(1/2)^k$.

   Answer: 31

   a. Identify the first included term, common ratio, and whether the sum is finite or infinite.
   b. Use $S_n = a(1 - r^n)/(1 - r)$ for a finite sum or $S = a/(1 - r)$ only when $|r| < 1$.
   c. Substitution and simplification give 31.

5. Evaluate $\sum_{k=0}^{\infty} 12(1/3)^k$.

   Answer: 18

   a. Identify the first included term, common ratio, and whether the sum is finite or infinite.
   b. Use $S_n = a(1 - r^n)/(1 - r)$ for a finite sum or $S = a/(1 - r)$ only when $|r| < 1$.
   c. Substitution and simplification give 18.

6. Evaluate $\sum_{k=0}^{\infty} 8(-1/2)^k$.

   Answer: $16/3$

   a. Identify the first included term, common ratio, and whether the sum is finite or infinite.
   b. Use $S_n = a(1 - r^n)/(1 - r)$ for a finite sum or $S = a/(1 - r)$ only when $|r| < 1$.
   c. Substitution and simplification give $16/3$.

7. Decide whether $\sum_{k=0}^{\infty} 5(1.02)^k$ converges.

   Answer: *diverges*

   a. Identify the first included term, common ratio, and whether the sum is finite or infinite.
   b. Use $S_n = a(1 - r^n)/(1 - r)$ for a finite sum or $S = a/(1 - r)$ only when $|r| < 1$.
   c. Substitution and simplification give *diverges*.

8. Evaluate $\sum_{n=3}^{\infty} 2(1/4)^n$.

   Answer: 1/24

   a. Identify the first included term, common ratio, and whether the sum is finite or infinite.
   b. Use $S_n = a(1 - r^n)/(1 - r)$ for a finite sum or $S = a/(1 - r)$ only when $|r| < 1$.
   c. Substitution and simplification give 1/24.

9. Rewrite $\sum_{n=1}^{\infty} 3^{1-n}$ in ar^k form and sum it.

   Answer: 3/2

   a. Identify the first included term, common ratio, and whether the sum is finite or infinite.
   b. Use $S_n = a(1 - r^n)/(1 - r)$ for a finite sum or $S = a/(1 - r)$ only when $|r| < 1$.
   c. Substitution and simplification give 3/2.

10. Write 0.333... as a fraction.

    Answer: 1/3

    a. Identify the first included term, common ratio, and whether the sum is finite or infinite.
    b. Use $S_n = a(1 - r^n)/(1 - r)$ for a finite sum or $S = a/(1 - r)$ only when $|r| < 1$.
    c. Substitution and simplification give 1/3.

11. Write 0.272727... as a fraction.

    Answer: 3/11

    a. Identify the first included term, common ratio, and whether the sum is finite or infinite.
    b. Use $S_n = a(1 - r^n)/(1 - r)$ for a finite sum or $S = a/(1 - r)$ only when $|r| < 1$.
    c. Substitution and simplification give 3/11.

12. Write 0.145145... as a fraction.

    Answer: 145/999

    a. Identify the first included term, common ratio, and whether the sum is finite or infinite.
    b. Use $S_n = a(1 - r^n)/(1 - r)$ for a finite sum or $S = a/(1 - r)$ only when $|r| < 1$.
    c. Substitution and simplification give 145/999.

13. A ball rebounds 70% of each previous height after a 10 m drop. Find total vertical distance.

    Answer: 170/3 m

    a. Identify the first included term, common ratio, and whether the sum is finite or infinite.
    b. Use $S_n = a(1 - r^n)/(1 - r)$ for a finite sum or $S = a/(1 - r)$ only when $|r| < 1$.

c. Substitution and simplification give 170/3 m.

14. Deposit 100 dollars at the end of each year for four years at 5 percent. Find the value immediately after the fourth deposit.

    Answer: 431.01 dollars

    a. Identify the first included term, common ratio, and whether the sum is finite or infinite.
    b. Use $S_n = a(1-r^n)/(1-r)$ for a finite sum or $S = a/(1-r)$ only when $|r| < 1$.
    c. Substitution and simplification give 431.01 dollars.

15. A square has area 1; each stage shades one fourth of the remaining area. Find total shaded area.

    Answer: 1

    a. Identify the first included term, common ratio, and whether the sum is finite or infinite.
    b. Use $S_n = a(1-r^n)/(1-r)$ for a finite sum or $S = a/(1-r)$ only when $|r| < 1$.
    c. Substitution and simplification give 1.

16. Find S_n for 7+7(0.8)+7(0.8) to the power 2+...

    Answer: $35(1-0.8^n)$

    a. Identify the first included term, common ratio, and whether the sum is finite or infinite.
    b. Use $S_n = a(1-r^n)/(1-r)$ for a finite sum or $S = a/(1-r)$ only when $|r| < 1$.
    c. Substitution and simplification give $35(1-0.8^n)$.

17. An infinite geometric series has first term 6 and sum 15. Find r.

    Answer: 3/5

    a. Identify the first included term, common ratio, and whether the sum is finite or infinite.
    b. Use $S_n = a(1-r^n)/(1-r)$ for a finite sum or $S = a/(1-r)$ only when $|r| < 1$.
    c. Substitution and simplification give 3/5.

18. An infinite geometric series has ratio -1/4 and sum 8. Find its first term.

    Answer: 10

    a. Identify the first included term, common ratio, and whether the sum is finite or infinite.
    b. Use $S_n = a(1-r^n)/(1-r)$ for a finite sum or $S = a/(1-r)$ only when $|r| < 1$.
    c. Substitution and simplification give 10.

19. A student uses a/(1-r) for r=2. Diagnose the error.

    Answer: $|r| \geq 1,$ so the infinite series diverges

    a. Identify the first included term, common ratio, and whether the sum is finite or infinite.
    b. Use $S_n = a(1-r^n)/(1-r)$ for a finite sum or $S = a/(1-r)$ only when $|r| < 1$.
    c. Substitution and simplification give $|r| \geq 1,$ so the infinite series diverges.

20. Evaluate $\sum_{n=2}^{6} 5(3)^{n-2}$.

    Answer: 605

a. Identify the first included term, common ratio, and whether the sum is finite or infinite.
b. Use $S_n = a(1 - r^n)/(1 - r)$ for a finite sum or $S = a/(1 - r)$ only when $|r| < 1$.
c. Substitution and simplification give 605.